MEDIUM · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Percentages

01

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 324

The correct answer is 324. It's given that 1,250 people attended the first information session, and that 72% of the people who attended the first information session attended the second information session. Therefore, the number of people who attended the first and second information sessions can be found by calculating 72% of 1,250, which is equal to $1,250\frac{72}{100}$, or 900. It's also given that 36% of the people who attended the first and second information sessions attended the third information session. Since 900 people attended the first and second information sessions, the number of people who attended the first, second, and third information sessions can be found by calculating 36% of 900, which is equal to $900\frac{36}{100}$, or 324. Therefore, 324 people attended all three information sessions.

Q2 110

The correct answer is 110. Let x represent the total number of items in the sample. It's given that 80% of the items are faulty and that there are 88 faulty items in the sample. Therefore, 80% of x is 88. Since 80% can be rewritten as $\frac{80}{100}$, it follows that $\frac{80}{100}x = 88$. Multiplying both sides of this equation by 100 yields $80x = 8,800$. Dividing both sides of this equation by 80 yields $x = 110$. Therefore, there are 110 total items in the sample.

Q3

C

C. 1.07

Choice C is correct. It's given that the population of City A increased by 7% from 2015 to 2016. Therefore, the population of City A in 2016 includes 100% of the population of City A in 2015 plus an additional 7% of the population of City A in 2015. This means that the population of City A in 2016 is 107% of the population in 2015. Thus, the population of City A in 2016 is $\frac{107}{100}$, or 1.07, times the 2015 population. Therefore, the value of k is 1.07.

Choice A is incorrect. This would be the value of k if the population in 2016 was 7% of the population in 2015.

Choice B is incorrect. This would be the value of k if the population in 2016 was 70% of the population in 2015.

Choice D is incorrect. This would be the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

Q4

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Answer not provided in source data — see rationale below.

Choice B is correct. If the stove's value depreciates by 20% of the previous year's estimated value, then each year it retains 100% – 20% = 80%, or 0.80, of the previous year's estimated value. Since the stove's value was \$800 when Thomas installed it, the estimated value after two years would be $(0.80)(0.80)(\$800) = \$512$.

Choice A is incorrect. This is the value of the stove if each year it had depreciated by 20% of the original value rather than by 20% of the previous year's estimated value. Choice C is incorrect and may be the result of a computational error. Choice D is incorrect. This is the estimated value of the stove 1 year after Thomas installed it, not 2 years.

Q5 C

C. 1,170,000

Choice C is correct. Of the 1,300,000 units sold during the first month, 15% were returned, so $(1{,}300{,}000)(0.15) = 195{,}000$ units were returned during the first month. If the units were sold and returned at the same rate for the next 5 months, then a total of $(195{,}000)(6) = 1{,}170{,}000$ smartphone units were returned during the 6-month period.

Choice A is incorrect. This is the number of units that were returned in 1 month. Choice B is incorrect. This is the number of units that were returned in 5 months. Choice D is incorrect. This is the number of units sold and not returned during the first 6 months.

Q6 B

B. 13

Choice B is correct. It's given that the value of z is 1.13 times 100. This can be written as $z = 1.13100$, which is equivalent to $z = 1 + 0.13(100)$, or $z = 1 + \frac{13}{100}100$. It follows that the value of z is 100% of 100 plus 13% of 100. Therefore, the value of z is 13% greater than 100.

Choice A is incorrect. This gives a value of z that is 1.113, not 1.13, times 100.

Choice C is incorrect. This gives a value of z that is 2.30, not 1.13, times 100.

Choice D is incorrect. This gives a value of z that is 3.13, not 1.13, times 100.

Q7 161

The correct answer is 161. For a number to be 40% greater than 115, it follows that the number is 100% of $115 + 40\%$ of 115, which can be written as $\frac{100}{100}115 + \frac{40}{100}115$. This expression is equivalent to $1115 + 0.4115$, or 1.4115, which is equal to 161. Therefore, 161 is 40% greater than 115.

Q8

A

A. $1.43w$

Choice A is correct. The result of increasing a positive quantity w by $x\%$ can be represented by the expression $1 + \frac{x}{100}w$. Therefore, the result of increasing a positive quantity w by 43% can be found by substituting 43 for x in the expression $1 + \frac{x}{100}w$, which gives $1 + \frac{43}{100}w$, or $1.43w$. Thus, the expression $1.43w$ represents the result of increasing a positive quantity w by 43%.

Choice B is incorrect. This is the result of decreasing a positive quantity w by 43%.

Choice C is incorrect. This is the result of increasing a positive quantity w by $4{,}200\%$.

Choice D is incorrect. This is the result of decreasing a positive quantity w by 57%.

Q9

184

The correct answer is 184. It's given in the table that there are 375 objects in group A and 690 objects in group C. It's also given that the number of objects in group C is $p\%$ of the number of objects in group A. Therefore, 690 is $p\%$ of 375, which can be represented by $690 = \frac{p}{100}375$, or $690 = 3.75p$. Dividing both sides of this equation by 3.75 yields $184 = p$. Therefore, the value of p is 184.

Q10

A

A. $1.09x$

Choice A is correct. Increasing the positive quantity x by 9% is the result of adding 9% of x to x. 9% of x can be represented algebraically as $\frac{9}{100}x$, or $0.09x$. Adding this expression to x yields $x + 0.09x$, or $1.09x$.

Choice B is incorrect. This represents 9% of x. Choice C is incorrect. This represents increasing x by 9, not by 9%. Choice D is incorrect. This represents increasing x by 0.09, not by 9%.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q4 — the source CB data omitted a structured key for these items, so the answer slot shows –. The worked solution in the rationale below each entry contains the correct value.

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